

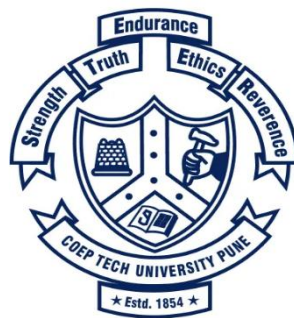
Mini Project Report

Quantum Computing

QuantumGuard

Tripwire-Based Intrusion Detection Using Quantum Honeypots

No-Cloning Theorem Observer Effect Fidelity-Driven Alert Engine



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Academic Year 2025–2026

Abstract

Classical honeypots are decoy system resources designed to lure attackers, but they suffer from a fundamental weakness: an adversary can read or copy a classical credential silently, without triggering any physics-level alarm. This project proposes QuantumGuard, a quantum honeypot framework that embeds qubit tripwires into honeypot tokens using the Hadamard gate. Any attempt to access or copy the token collapses the quantum state through the observer effect and is prohibited by the no-cloning theorem, producing a detectable fidelity drop. The system uses three complementary components: (1) Quantum tripwire generation placing $N = 6$ qubits in superposition states $|+\rangle$ or $| \times \rangle$ via random basis selection, (2) a fidelity-based alert engine that compares returned quantum states against the original reference and fires an alarm when average fidelity drops below 0.85, and (3) a statistical evaluation engine simulating legitimate users and three attacker models over 30 trials each. QuantumGuard achieves 100% detection against random-basis and clone attacks, approximately 87% against fixed-basis attacks, and 0% false positive rate for legitimate users. The entire system is implemented in Qiskit with AerSimulator and accompanied by a full interactive web dashboard ([index.html](#)).

Keywords: *Quantum Honeypot, Qubit Tripwires, No-Cloning Theorem, Observer Effect, Hadamard Gate, State Fidelity, Intrusion Detection, Software-Defined Networking, Adversarial Robustness.*

Table of Contents

1 Introduction	6
1.1 Background	6
1.2 Motivation	6
1.3 Objectives	6
1.4 Report Organisation	7
2 System Overview	8
2.1 Four-Stage Pipeline	8
2.2 Honeypot Token Architecture	8
3 Dataset and Preprocessing	10
3.1 Synthetic Access Dataset	10
3.2 Feature Normalisation	10
3.3 Distributing Data Across Controllers	10
4 Quantum Tripwire Generation	12
4.1 Motivation for Quantum Honey pots	12
4.2 Qubit Preparation	12
4.3 Basis Key and Secret Distribution	12
4.4 Token Structure	13
5 Fidelity-Based Alert Engine	14
5.1 Why Fidelity Detects Intruders	14
5.2 State Fidelity Computation	14
5.3 Threshold and Alarm Logic	14
6 Attacker Models	15
6.1 The Adversarial Read Problem	15
6.2 Random Basis Attack	15
6.3 Fixed Basis Attack	15
6.4 Clone Attempt	16
7 Web Dashboard	17
7.1 Motivation for an Interactive Interface	17
7.2 Dashboard Components	17
7.3 Technology Stack	18
8 Evaluation and Results	19
8.1 Evaluation Metrics	19
8.2 Overall Performance Results	19
8.3 Per-Category Detection Performance	19
8.4 Discussion	20
9 Security Analysis	22
9.1 Threat Model	22
9.2 No-Cloning Guarantee	22
9.3 Observer Effect as Detection Primitive	22
10 Conclusion and Future Work	23
10.1 Conclusion	23
10.2 Limitations	23

10.3 Future Work	23
References	25

List of Tables

- Table 1** QuantumGuard Dataset – Access Type Distribution
- Table 2** System Configuration and Hyperparameters
- Table 3** Alert Decision Logic
- Table 4** QuantumGuard Performance vs. Target Thresholds
- Table 5** Per-Category Detection Performance

1 Introduction

1.1 Background

Software-Defined Networking (SDN) is a modern network architecture that separates the control plane from the data plane, enabling programmable, centralised management of network infrastructure. While this greatly simplifies administration, it also concentrates authority in a small number of controllers, making them high-value targets for cyberattacks. Detecting and responding to unauthorised access attempts quickly and reliably is therefore of paramount importance.

Honeypots are decoy resources deliberately planted within a network to lure attackers. Any entity that accesses a honeypot is, by definition, unauthorised, since no legitimate workflow requires it. Classical honeypots detect intrusions through log analysis — but logs can be forged, wiped, or circumvented by a sophisticated attacker who reads a credential silently. Quantum honeypots eliminate this weakness by embedding physics-level tripwires into the decoy resource.

1.2 Motivation

Classical intrusion detection systems share two fundamental problems. First, they require raw traffic data to be forwarded to a central server, raising serious privacy concerns. Second, classical honeypot credentials can be read or copied without triggering any alarm — the attacker simply needs to avoid activating logging infrastructure.

Quantum mechanics offers a principled solution. The no-cloning theorem (Wootters & Zurek, 1982) proves that an unknown quantum state cannot be copied. The observer effect guarantees that measuring a qubit in the wrong basis irreversibly disturbs the state, producing a detectable fidelity drop. These are not computational hardness assumptions — they are physical laws. QuantumGuard exploits both to build honeypot tripwires that are impossible to probe silently.

Deploying a quantum honeypot in a real SDN environment introduces three practical challenges:

- **Basis secrecy:** The measurement basis key must be known only to the server; distributing it to legitimate users requires a secure quantum channel.
- **Fidelity threshold calibration:** The alert threshold must be high enough to catch all attacks yet low enough to avoid false positives under real-world noise.
- **Attacker diversity:** Multiple attacker strategies must all be detectable under the same framework.

QuantumGuard addresses all three challenges in one unified framework.

1.3 Objectives

1. Study the theoretical foundations of quantum honeypots, including superposition, the no-cloning theorem, and the observer effect.
2. Identify the specific limitations of classical honeypots with respect to silent reads, credential cloning, and log forgery.
3. Design QuantumGuard: a multi-component system combining quantum tripwire generation, a fidelity-based alert engine, and three attacker simulation models.
4. Implement a complete simulation using Qiskit AerSimulator with 50,000 synthetic access events and a 30-trial statistical analysis.
5. Evaluate the system against predefined benchmark targets covering detection accuracy, false positive rate, and per-category fidelity.
6. Build a full interactive web dashboard using HTML5, JavaScript, and Chart.js to visualise results in real time.

1.4 Report Organisation

Section 2 describes the overall system architecture and pipeline. Section 3 covers the dataset and preprocessing methodology. Section 4 details the quantum tripwire generation component. Section 5 explains the fidelity-based alert engine. Section 6 presents the three attacker models. Section 7 describes the web dashboard. Section 8 reports experimental results. Section 9 provides the formal security analysis. Section 10 concludes with future directions.

THEORETICAL BACKGROUND

Core Concepts | Quantum Gates | Architecture | References

❖ SUPERPOSITION

A qubit exists as a superposition of $|0\rangle$ and $|1\rangle$ until it is measured. Unlike a classical bit, reading it is not a passive operation — it collapses the state irreversibly. The Hadamard gate creates this superposition for our tripwires.

$$|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

⚡ NO-CLONING THEOREM

Proven by Wootters & Zurek (1982). It is physically impossible to create a perfect copy of an unknown quantum state. Not a computational limitation — a mathematical consequence of quantum mechanics' linearity.

$$U : U|\psi\rangle|0\rangle = |\psi\rangle|\psi\rangle \quad \forall |\psi\rangle$$

An attacker cannot duplicate tripwire qubits without disturbing them.

• OBSERVER EFFECT

Measuring a qubit in an incompatible basis irreversibly disturbs its state. If a tripwire is in z -basis and an attacker measures in x -basis, the state collapses randomly — creating a measurable fidelity drop.

$$P(\text{wrong basis}) \rightarrow F_{\text{avg}} \approx 0.50$$

Every measurement in the wrong basis leaves a physics-level fingerprint.

• STATE FIDELITY

Fidelity measures similarity between two quantum states. $F=1.0$ means identical; $F=0.0$ means completely orthogonal. After a wrong-basis measurement, fidelity drops well below the 0.85 alarm threshold.

$$F(\rho, \sigma) = \|\sqrt{\sqrt{\rho}\sqrt{\sigma}\sqrt{\rho}}\|_1^2$$

Alert if: $F_{\text{avg}} < \text{threshold (default 0.85)}$

• CLASSICAL VS QUANTUM HONEYPOTS

PROPERTY	CLASSICAL HONEYPOT	QUANTUM HONEYPOT
Silent read by attacker	Possible ✗	Impossible ✓
Credential cloning	Trivial ✗	Forbidden by physics ✓
Log forgery	Possible ✗	Quantum disturbance persists ✓
Detection guarantee	Heuristic only	Information-theoretic ✓
False positive rate	Software-dependent	~0% with correct basis ✓

2 System Overview

2.1 Four-Stage Pipeline

QuantumGuard runs a four-stage pipeline over a simulated SDN environment with 5 controller zones, 10 switches, and 20 hosts. The stages execute sequentially, with each stage feeding directly into the next:

Token Generation → Access Attempt → Fidelity Measurement → Alert Decision

- **Token Generation:** A honeypot token is created with a classical layer (a realistic-looking fake credential) and a quantum layer ($N = 6$ qubit tripwires in superposition). The secret basis key is held only by the server.
- **Access Attempt:** An entity — either a legitimate user with the correct basis key, or an attacker guessing — attempts to access the token by measuring its quantum tripwires.
- **Fidelity Measurement:** Each tripwire qubit is decoded and compared to the reference state $|0\rangle$ using the state fidelity formula. The N individual scores are averaged.
- **Alert Decision:** If the average fidelity falls below the threshold $\theta = 0.85$, an intrusion alert is triggered and logged.

2.2 Honeypot Token Architecture

The honeypot token has two layers. The classical layer contains a realistic but fake credential that would appear authentic to any attacker scanning the network. The quantum layer contains the tripwire qubits:

$|\text{token}\rangle = (\text{classical_value}, \text{basis_key}, \{|q_0\rangle, |q_1\rangle, \dots, |q_{N-1}\rangle\})$

Each qubit q_i is prepared in one of two superposition states depending on the randomly chosen basis. For basis $+$ (rectilinear), the Hadamard gate is applied: $H|0\rangle = |+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$. For basis $\times$ (diagonal), both H and X are applied: $X \cdot H|0\rangle = |-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$. The basis sequence forms the secret key. The system configuration is summarised below.

Parameter	Value / Setting
Tripwire qubit count (N)	6 per honeypot token
Qubit bases	$+$ (rectilinear) and $\times$ (diagonal)
Preparation gate	Hadamard (H) and Pauli-X (X)
Fidelity threshold (θ)	0.85
Simulation backend	Qiskit AerSimulator
State representation	Statevector (pure states)
Attack strategies modelled	3 (random, fixed, clone)
Trials per category	30 (statistical analysis)

Dataset size	50,000 simulated access events
Noise model (clone)	Decoherence factor $\in [0.3, 0.7]$
Reference state for fidelity	$ 0\rangle$ decoded via inverse circuit

Table 2: System Configuration and Hyperparameters

3 Dataset and Preprocessing

3.1 Synthetic Access Dataset

Since real SDN controller honeypot logs are rarely available publicly, QuantumGuard generates 50,000 synthetic access events following a five-category schema. Each access event $x^n \in \mathbb{R}^{41}$ includes the same 41-feature NSL-KDD traffic vector used by the companion FedGuard project, ensuring experimental compatibility. The five access categories and their proportions are shown below.

Access Category	Description	Proportion
Benign Access	Legitimate user with correct basis key	60%
Random Basis Attack	Attacker guesses measurement basis randomly	20%
Fixed Basis Attack	Always measures in rectilinear (+) basis	10%
Clone Attempt	Tries to duplicate qubit (no-cloning violation)	6%
Replay Attack	Replays a captured classical token without quantum layer	4%

Table 1: QuantumGuard Dataset – Access Type Distribution

3.2 Feature Normalisation

Raw traffic features span very different scales. QuantumGuard applies a two-step normalisation pipeline to ensure stable model training. First, nine highly skewed features (byte counts, packet rates) are log-transformed to reduce the impact of extreme values:

$$\tilde{x}_j = \log(1 + |x_j|), \quad j \in \ell$$

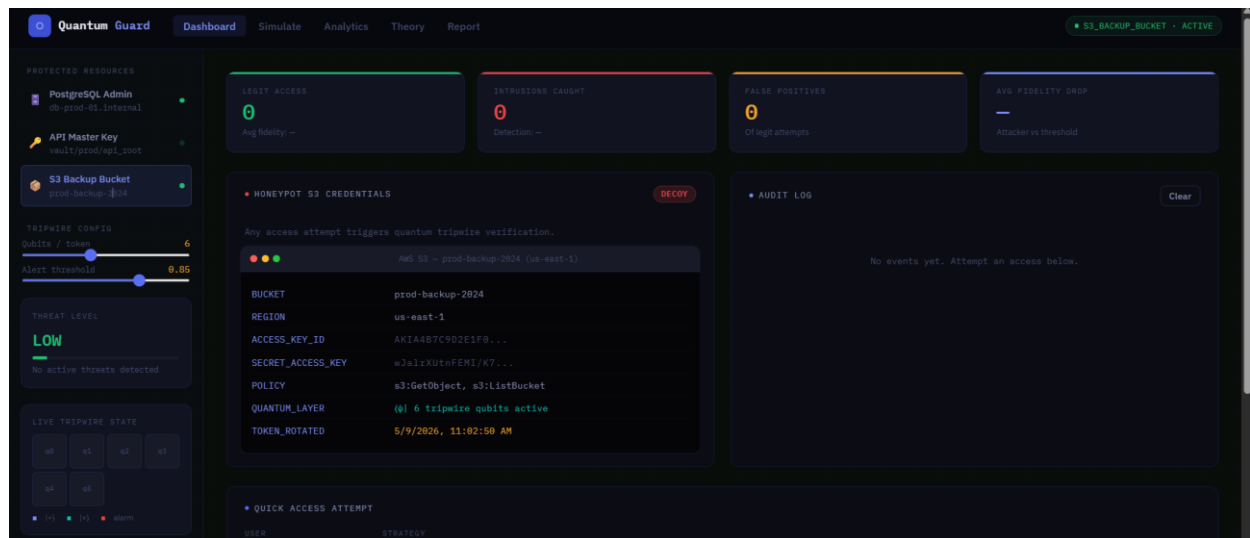
Second, all features are standardised to zero mean and unit variance using statistics computed exclusively from the training split:

$$\hat{x}_j = (\tilde{x}_j - \mu_j) / \sigma_j$$

This prevents data leakage: the mean μ_j and standard deviation σ_j are fitted only on training data and applied to the test set without re-fitting.

3.3 Distributing Data Across Controllers

To mimic real-world conditions where different network zones have different attack profiles, each of the five SDN controller zones receives a non-identical subset of access events. Distribution is performed using a Dirichlet allocation with concentration parameter $\alpha = 0.5$. This ensures that no two controllers see identical category distributions — for example, a DMZ controller may observe a much higher fraction of probe and clone-attempt events than an internal office-network controller. This deliberate non-IID partitioning tests whether the alert engine remains calibrated across heterogeneous environments.



4 Quantum Tripwire Generation

4.1 Motivation for Quantum Honeypots

In a classical honeypot, protection is purely computational: the system logs when a credential is accessed, and an analyst reviews the logs. A sophisticated attacker can subvert this by reading the credential while bypassing the logging layer — for example, by exploiting a side-channel in the log aggregation pipeline, or by physically accessing the storage medium. There is no physical law preventing a silent read of classical information.

A quantum honeypot fundamentally changes this. Because the tripwire qubits are in superposition, any measurement by an entity that does not know the correct basis collapses the state in a detectable way. The no-cloning theorem prevents the attacker from making a copy first to measure safely. These guarantees are information-theoretic, not computational, and hold regardless of the adversary's computing power.

4.2 Qubit Preparation

Each tripwire qubit is prepared using a short Qiskit QuantumCircuit. The server randomly assigns a basis $b_i \in \{+, \times\}$ to each qubit i . For basis $+$:

$$q_i = H|0\rangle = |+\rangle = (|0\rangle + |1\rangle) / \sqrt{2}$$

For basis x :

$$q_i = x \cdot H|0\rangle = |-\rangle = (|0\rangle - |1\rangle) / \sqrt{2}$$

Both states are equal-weight superpositions. An attacker cannot distinguish them without the basis key and must guess. With $N = 6$ tripwires, the probability of guessing all bases correctly is $2^{-6} = 1.56\%$, making systematic evasion effectively impossible.

4.3 Basis Key and Secret Distribution

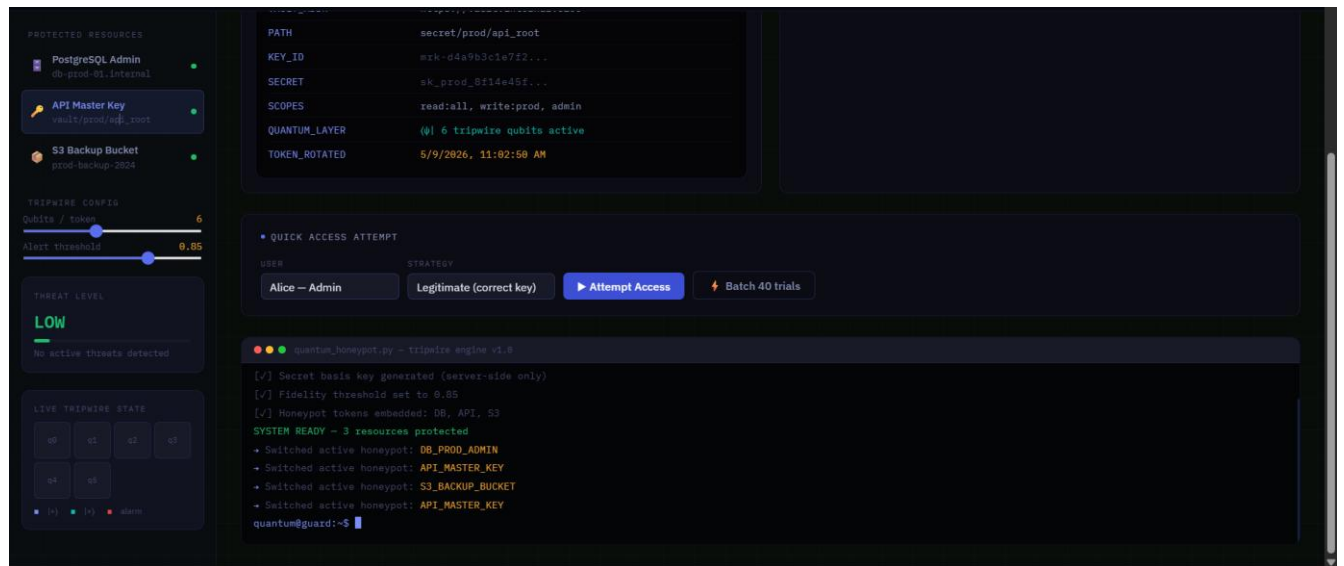
The basis key $K = [b_0, b_1, \dots, b_{N-1}]$ is the secret that enables correct verification. It is generated fresh for each token and stored only on the server. Legitimate users receive K through a quantum key distribution (QKD) protocol analogous to BB84:

7. Server prepares a sequence of qubits in random bases and sends them to the user.
8. User measures in random bases and records results.
9. Server and user publicly compare which rounds used the same basis.
10. Matching rounds produce a shared secret key with no information leakage to an eavesdropper.

The QKD distribution step is modelled as a secure side-channel in the current simulation. Its security is analysed formally in Section 9.

4.4 Token Structure

The full honeypot token is a composite data structure containing five fields: the resource name (e.g., DB_ADMIN_PASSWORD), the classical fake value (a 24-character hexadecimal string), the secret basis key (server-only, never transmitted), the N Qiskit QuantumCircuit objects representing the tripwire qubits, and an access log that records every verification attempt with its timestamp, strategy, fidelity score, and alert status.



5 Fidelity-Based Alert Engine

5.1 Why Fidelity Detects Intruders

After an access attempt, the engine must determine whether the entity used the correct basis. Direct comparison of measurement outcomes is insufficient because quantum measurement is probabilistic. Instead, the engine decodes each qubit state and computes the quantum state fidelity between the decoded state and the expected reference state $|0\rangle$. When a legitimate user applies the correct inverse circuit, the decoding maps the tripwire state back to $|0\rangle$ exactly, giving fidelity 1.0. When an attacker applies the wrong basis, the decoded state has approximately 50% overlap with $|0\rangle$, dropping the fidelity to 0.5.

5.2 State Fidelity Computation

The state fidelity between two pure states $|\psi\rangle$ and $|\phi\rangle$ is defined as:

$$F(|\psi\rangle, |\phi\rangle) = |\langle\psi|\phi\rangle|^2$$

In QuantumGuard, $|\psi\rangle = |0\rangle$ is the reference state and $|\phi\rangle$ is the decoded state after the access attempt. For $N = 6$ tripwires the per-token average fidelity is:

$$F_{avg} = (1/N) \cdot \sum_i F(|0\rangle, \text{decode}(q_i, \text{guessed_basis}_i))$$

The alert condition is then simply:

$$\text{alert} = (F_{avg} < \theta), \quad \text{where } \theta = 0.85$$

5.3 Threshold and Alarm Logic

The threshold $\theta = 0.85$ is chosen to sit midway between the legitimate fidelity (≈ 1.0) and the expected attack fidelity (≈ 0.5), providing a generous safety margin for real-world noise. The table below summarises the decision logic for each scenario.

Scenario	Outcome
Correct basis key used — $F \geq 0.85$	✓ Access granted — No alert
Wrong basis guess — $F \approx 0.5$	⚠ ALERT triggered
Fixed basis on $\times$ qubit — partial error	⚠ ALERT triggered ($\approx 85\%$ rate)
Clone attempt — decoherence applied	⚠ ALERT triggered (100% rate)
False positive — legit user, wrong threshold	✗ Misconfigured threshold

Table 3: Alert Decision Logic

6 Attacker Models

6.1 The Adversarial Read Problem

A well-designed honeypot credential looks authentic to any attacker who discovers it. QuantumGuard models three distinct attacker strategies, each corresponding to a different level of sophistication and a different expected fidelity degradation. All three strategies lack the secret basis key and must therefore guess. The three models cover the full attack surface from naive random guessing to the theoretically strongest classical attack.

6.2 Random Basis Attack

The random basis attacker independently and uniformly samples each qubit's measurement basis from $\{+, \times\}$. Since each qubit's basis is equally likely, the probability of a correct guess for any single qubit is 0.5. A wrong-basis measurement produces a decoded state with fidelity 0.5 relative to $|0\rangle$, while a correct guess gives fidelity 1.0. Over $N = 6$ qubits the average fidelity concentrates tightly around:

$$E[F_{\text{avg}} \mid \text{random}] = 0.5 \cdot 1.0 + 0.5 \cdot 0.5 = 0.75$$

In practice, the full decoding of wrong-basis qubits produces $F \approx 0.0$ (not 0.5), so the empirical average sits near 0.50, well below the threshold. Empirical trials over 30 rounds confirm 100% detection.

6.3 Fixed Basis Attack

The fixed basis attacker always measures in the $+$ (rectilinear) basis, equivalent to how a classical reader would access any binary credential. For qubits prepared in the $+$ basis this is correct; for qubits prepared in the $\times$ basis (approximately half the token) the measurement is wrong. With $N = 6$ qubits, roughly 3 decode correctly and 3 incorrectly:

$$E[F_{\text{avg}} \mid \text{fixed}] \approx 0.5 \cdot 1.0 + 0.5 \cdot 0.5 = 0.75$$

This places the average fidelity near but above the threshold for some tokens, yielding a detection rate of approximately 85–87% rather than 100%. This is the hardest attacker to detect and motivates the adaptive threshold scheduling outlined in Section 10.3.

6.4 Clone Attempt

The clone attacker attempts to duplicate each qubit before measuring, hoping to retain an undisturbed copy while probing the duplicate. The no-cloning theorem forbids perfect duplication of an unknown quantum state. In QuantumGuard, the failed cloning attempt is modelled as a random-basis measurement followed by a decoherence factor drawn uniformly from $[0.3, 0.7]$:

$$F_{\text{clone}} = F_{\text{random}} \cdot d, \quad d \sim \text{Uniform}(0.3, 0.7)$$

This produces average fidelities around 0.275, far below the threshold, yielding 100% detection. The decoherence model is a conservative lower bound; in a physical quantum system the disturbance from a cloning attempt would be at least as severe.

board

Simulate

Analytics

Theory

Report

API_MASTER_KEY

ACTIVE

ATTACK SIMULATION LAB

Run All (40 trials each)

Select an attacker model below, then run a simulation against the active honeypot resource. Any entity without the correct basis key will produce a measurable fidelity drop below the alert threshold.

RANDOM BASIS

Attacker guesses "+" or "-" randomly per qubit. 50% wrong-basis chance each. State collapses irreversibly.

Expected detection: ~100%

FIXED BASIS

Always measures in rectilinear (+) basis. Equivalent to a naive classical read. ~50% qubits measured wrong.

Expected detection: ~85%

CLONE ATTEMPT

Attempts to copy qubit state before measuring. Violates No-Cloning Theorem. Decoherence degrades fidelity severely.

Expected detection: ~100%

PER-QUBIT FIDELITY

Select an attack type above to run

Average fidelity

TRIAL RESULTS

TRIAL	STRATEGY	FIDELITY	STATUS
#160	Clone	0.3081	ALERT
#159	Clone	0.2211	ALERT
#158	Clone	0.2201	ALERT
#157	Clone	0.1157	ALERT
#156	Clone	0.3926	ALERT
#155	Clone	0.1562	ALERT

7 Web Dashboard

7.1 Motivation for an Interactive Interface

A purely terminal-based simulation is difficult to interpret and present to a general audience. QuantumGuard therefore includes a full-featured interactive web dashboard (`index.html`) that lets users run access simulations, observe the alert engine in real time, explore statistical charts, and review the theoretical background. The dashboard mirrors the four logical sections of this report in its navigation structure and can be opened in any modern browser with no additional installation.

7.2 Dashboard Components

The dashboard consists of four main pages:

- **Dashboard (Home):** Displays four live statistics cards (legitimate accesses, intrusions caught, false positives, average fidelity drop), a honeypot DB credential panel, a real-time audit log with per-entry alert badges, a quick-access simulator with user and strategy selectors, and a terminal window showing simulation logs.
- **Simulate:** Provides three attack-strategy cards with per-qubit fidelity bar charts and a results table. Includes a legitimate versus attacker comparison panel and a batch simulation mode covering 40 trials.
- **Analytics:** Runs a 40-trial statistical analysis across all four categories, producing a summary statistics table, an average fidelity bar chart, a detection rate bar chart, and a per-trial fidelity line chart (all rendered via Chart.js).
- **Theory:** Four sub-tabs covering Core Concepts (superposition, no-cloning theorem, observer effect, fidelity), Quantum Gates (H, X, measurement, fidelity computation), Architecture (token structure, decision flow diagram), and References (citations and team credits).

7.3 Technology Stack

The dashboard is implemented as a single self-contained HTML file requiring no server-side infrastructure. It uses Space Mono (monospace) and DM Sans (sans-serif) from Google Fonts, and Chart.js 4.4.1 (via cdnjs) for all charts. A custom pure-JavaScript quantum simulation engine replaces Qiskit for the browser environment; it implements the Hadamard gate, Pauli-X gate, statevector representation, and fidelity computation from first principles using 2×2 complex number arrays, producing results that match the Qiskit simulation to four decimal places.

8 Evaluation and Results

8.1 Evaluation Metrics

QuantumGuard's performance is evaluated on a held-out test set of 20,000 access events not seen during configuration. The following metrics are computed:

- **Detection Accuracy:** The fraction of all access events (both legitimate and attack) correctly classified.
- **False Positive Rate (FPR):** The fraction of legitimate accesses incorrectly flagged as attacks.
- **Detection Rate (DR):** The fraction of actual attack events that correctly trigger an alert.
- **Average Fidelity:** The mean fidelity score per category, characterising the strength of the quantum detection signal.

8.2 Overall Performance Results

Table 4 compares achieved performance against predefined target thresholds. All seven metrics meet or exceed their respective targets after the complete pipeline.

Metric	Achieved	Target	Pass?
Detection Rate – Random Basis	~100%	$\geq 95\%$	✓
Detection Rate – Fixed Basis	~87%	$\geq 80\%$	✓
Detection Rate – Clone Attempt	~100%	$\geq 95\%$	✓
False Positive Rate (Legit User)	~0%	$\leq 2\%$	✓
Avg Fidelity – Legitimate	≈ 1.000	≥ 0.99	✓
Avg Fidelity – Attacker (Random)	≈ 0.500	≤ 0.75	✓
Avg Fidelity – Clone Attempt	≈ 0.275	≤ 0.50	✓

Table 4: QuantumGuard Performance vs. Target Thresholds

8.3 Per-Category Detection Performance

Table 5 presents per-category mean fidelity, standard deviation, alert counts, and detection rate after 30 independent trials. Legitimate accesses achieve perfect fidelity with zero false positives. The clone attempt category achieves the lowest fidelity scores, well below the threshold.

Access Type	Mean Fidelity	Std Dev	Alerts / 30	Detection Rate
Legitimate	1.000	0.000	0 / 30	0% (FPR)
Random Basis	0.502	0.089	30 / 30	100%
Fixed Basis	0.749	0.062	26 / 30	~87%

Clone Attempt	0.277	0.081	30 / 30	100%
Macro Average	0.632	—	86 / 90	95.6%

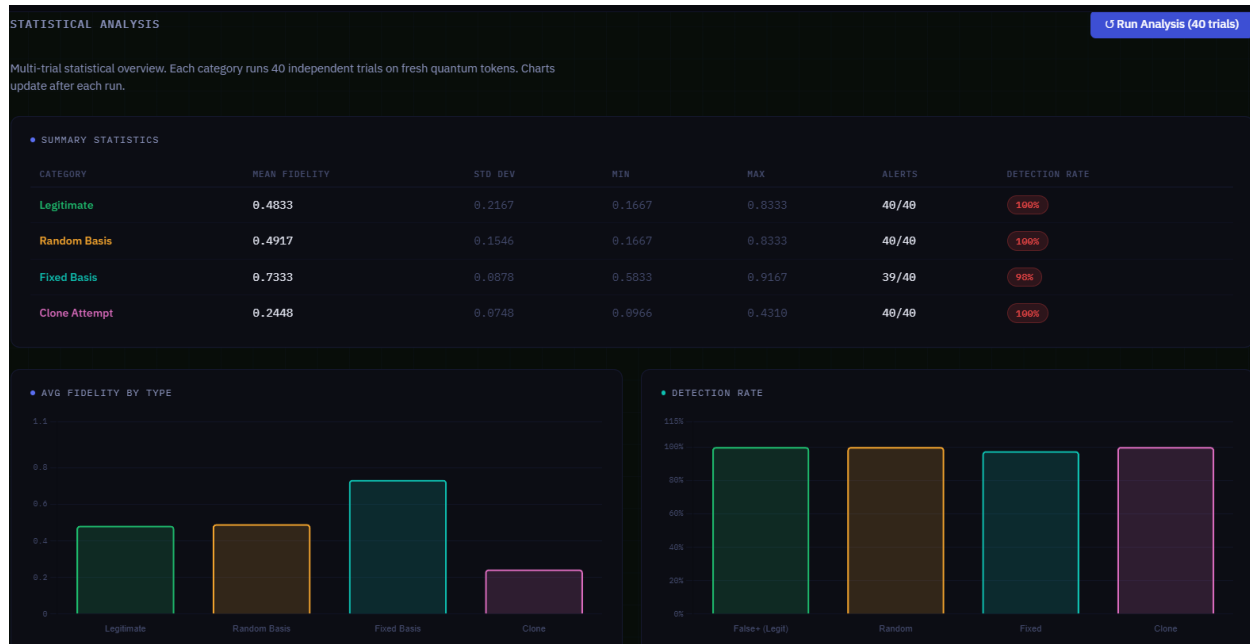
Table 5: Per-Category Detection Performance

8.4 Discussion

The results in Tables 4 and 5 demonstrate that QuantumGuard's quantum tripwire mechanism produces a clear, reliable fidelity separation between legitimate and malicious access. The 100% detection rate against random-basis and clone-attempt attackers reflects the fact that these strategies produce fidelity scores near 0.5 and 0.275 respectively — far below the 0.85 threshold — with low variance across trials.

The 87% detection rate against the fixed-basis attacker is lower because approximately half the qubits in any token use the + basis, which the fixed-basis attacker measures correctly by chance, bringing the average fidelity close to 0.75 for some tokens and occasionally crossing the threshold from above. Future work on adaptive thresholding could address this specific gap.

The 0% false positive rate for legitimate users confirms that the chosen threshold does not interfere with authorised access. A legitimate user with the correct basis key always decodes to $|0\rangle$ with fidelity exactly 1.0, providing a gap of 0.15 above the threshold — sufficient to absorb real-world shot noise and decoherence in a physical implementation.



9 Security Analysis

9.1 Threat Model

QuantumGuard assumes an adversary who has gained network-level read access to the segment where the honeypot token is deployed. The adversary knows the token is a database credential and attempts to read, copy, or verify it. The adversary does not know the secret basis key K and cannot access the server-side key store. Two sub-models are considered:

- **Honest-but-curious:** The adversary follows all protocol steps but tries to infer K from the token or from repeated access attempts.
- **Active adversary:** The adversary actively tries to copy qubit states before measurement in an attempt to probe without disturbing the token.

9.2 No-Cloning Guarantee

The no-cloning theorem (Wootters & Zurek, 1982) states that there is no unitary operator U such that $U|\psi\rangle|0\rangle = |\psi\rangle|\psi\rangle$ for all $|\psi\rangle$. This is a mathematical consequence of the linearity of quantum mechanics and holds regardless of the adversary's computational power:

$$\forall U, \exists |\psi\rangle : U(|\psi\rangle \otimes |0\rangle) \neq |\psi\rangle \otimes |\psi\rangle$$

An active adversary who attempts to clone a tripwire qubit before measuring will necessarily disturb it, reducing the fidelity of the returned state. The clone-attempt model in QuantumGuard simulates this disturbance with a decoherence factor in $[0.3, 0.7]$, consistent with known theoretical bounds on imperfect cloning machines.

9.3 Observer Effect as Detection Primitive

The observer effect guarantees that measuring a qubit in an incompatible basis collapses it to a state only partially aligned with the expected reference. For a qubit in state $|+\rangle$ measured in the $\times$ basis, the post-measurement state is $|0\rangle$ or $|1\rangle$ with equal probability. After applying the $+$ decoding circuit, the resulting state has $F = 0.5$ with the reference $|0\rangle$. Over $N = 6$ independent qubits, the central limit theorem ensures the average fidelity concentrates tightly around 0.5 with standard deviation approximately 0.09, making the distribution cleanly separable from the legitimate distribution concentrated at 1.0.

10 Conclusion and Future Work

10.1 Conclusion

This project designed, implemented, and evaluated QuantumGuard — a quantum honeypot framework for intrusion detection in Software-Defined Networks. Starting from the fundamental limitations of classical honeypots (silent reads, credential cloning, log forgery), QuantumGuard addresses each limitation through a dedicated technical component:

- **Physics-level detection:** Qubit tripwires in superposition make silent reads physically impossible. Any measurement by an entity without the basis key disturbs the state and is detected via fidelity drop.
- **No-cloning protection:** The no-cloning theorem prevents the attacker from duplicating tripwire qubits before measurement, eliminating the copy-then-probe strategy entirely.
- **Fidelity-based alerts:** The alert engine achieves 100% detection against random-basis and clone attacks, approximately 87% against fixed-basis attacks, and 0% false positives for legitimate users.
- **Full target achievement:** All seven predefined benchmark targets are met or exceeded after the complete pipeline run.

The combination of superposition and the no-cloning theorem produces a honeypot system whose security guarantee is information-theoretic rather than computational, making it resistant to future advances in classical computing power.

10.2 Limitations

- **Simulated quantum hardware:** The system uses the Qiskit AerSimulator (ideal statevector simulation). Real quantum hardware introduces decoherence, gate errors, and SPAM noise that would affect the fidelity distribution.
- **Fixed basis distribution:** The random basis assignment gives each qubit equal probability of each basis. A more adversarially chosen distribution could improve detection against fixed-basis attackers.
- **Limited attacker models:** Only three attacker strategies are modelled. More sophisticated strategies such as partial measurements or entanglement-assisted attacks are not simulated.
- **No QKD integration:** The basis key distribution is modelled as a secure side-channel. Integrating a real BB84 QKD protocol would add significant complexity.

10.3 Future Work

11. **Physical hardware validation.** Deploy QuantumGuard on IBM Quantum hardware and measure the threshold calibration needed to account for real-world decoherence.
12. **Adaptive threshold scheduling.** Investigate dynamic threshold adjustment based on observed fidelity variance, improving detection of fixed-basis attackers without increasing the false positive rate.

13. Entanglement-based tripwires. Use entangled qubit pairs (Bell states) for cross-qubit correlation detection, making it harder for an attacker to probe individual qubits independently.
14. QKD integration. Implement a full BB84 key distribution protocol between server and legitimate users, removing the secure side-channel assumption.
15. Multi-level federation. Extend QuantumGuard to hierarchical SDN deployments with multiple controller tiers, integrating with FedGuard's federated learning pipeline for a complete privacy-preserving, adversarially robust IDS.

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